AI-Powered Early Disease Risk Prediction & Remote Patient Monitoring System

Capstone Project — Phase I: Project Initiation

Problem Statement

Chronic metabolic conditions such as diabetes are often diagnosed late because monitoring typically occurs only during infrequent clinical visits, relying on generic population-level risk thresholds. This delays early intervention and increases the likelihood of complications that could otherwise be prevented or managed proactively.

Proposed Solution

A system that continuously monitors patient vitals and lifestyle indicators using wearable/IoT sensors, applies machine learning to predict early diabetes and metabolic risk, and delivers real-time, explainable alerts to both patients and clinicians through a connected dashboard and mobile app.

Objectives

- Build a predictive model that estimates diabetes/metabolic risk from patient vitals and lifestyle data.
- Enable continuous remote monitoring rather than one-time clinical snapshots.
- Generate real-time alerts for both patients and clinicians.
- Make predictions explainable, so users understand the contributing factors behind a risk flag.

Scope

In scope (this phase): diabetes and pre-diabetic risk prediction using structured vitals and lifestyle datasets, a working ML pipeline, and a monitoring dashboard prototype.
Out of scope (future work): multi-disease expansion and full clinical-grade hardware integration, planned beyond this capstone phase.

Novelty

- Personalized baselines — the model learns each patient's individual normal range instead of applying generic population thresholds.
- Explainable AI — every risk alert is accompanied by its contributing factors, supporting clinical trust rather than a black-box score.

Feasibility

Relevant datasets have been identified and secured, and initial exploratory data analysis is complete. The current phase focuses on model training and evaluation, followed by dashboard integration — a realistic and achievable path within the project timeline.

Major Components / Technologies

Layer	Description
Data layer	Diabetes datasets and simulated wearable/sensor streams
ML layer	Risk-prediction model (e.g., Random Forest / XGBoost / LSTM)
Backend / API layer	Processing pipeline, risk scoring, alert generation
Alert & notification engine	Severity tiering and routing of real-time alerts
Frontend	Doctor dashboard and patient mobile app

Alert & Notification Flow

1. **Risk scoring** — new patient data (glucose reading, wearable batch, etc.) is scored by the ML model, producing a probability plus contributing factors.
2. **Threshold + personalized baseline check** — the score is compared against both a general clinical threshold and the patient's own learned baseline.
3. **Severity tiering** — alerts are bucketed as informational, moderate, or urgent.
4. **Routing** — urgent/moderate alerts go to the doctor dashboard with trend and explainability detail; relevant alerts are simplified for the patient app.
5. **Logging** — every alert is logged with its triggering data snapshot for review and model performance auditing.

Expected Outcome

A working prototype that predicts diabetes risk early and supports remote, continuous patient monitoring — helping shift care from reactive to preventive.

Video Notes

Suggested duration: 3–5 minutes. Must feature team members' faces and voices, be approved by the project guide before submission, and disclose the AI tools used in production.